

Subject Code	ENGL5036
Subject Title	Corpus Linguistics and AI
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	None
Objectives	1. To understand concepts and issues in the field of corpus linguistics. 2. To explore how corpus linguistics, as one of the first areas in linguistics to work with large scale language data, contributed to the development of large language models and artificial intelligence. 3. To learn how to use corpora (large collections of electronic text) for linguistic studies. 4. To learn how to use computer-assisted corpus tools and AI to discover patterns of language structure, use and meaning in different corpora. 5. To study how corpus linguistics can contribute to our understanding of language. 6. To study different applications of corpus linguistics, e.g. dictionaries and grammars, register variation, language education and research, translation, critical language study, development of large language models, etc. 7. To explore the historical and contemporary interfaces between corpus linguistics and artificial intelligence (e.g. natural language processing). To discuss how current and future corpus linguistics work can leverage maturing AI technologies in various aspects, for example methodological (e.g. generating, collecting, analyzing, tagging data, etc), or expanding the scope of the field (e.g. multimodal data).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills a) Apply corpus linguistic and AI techniques to different types of corpus as databases to study a wide range of linguistic features; b) establish a direct link between theories about language and what is observed in natural language; c) utilize artificial intelligence effectively in conducting corpus-based and corpus-driven research projects d) report, in the form of a presentation and a written report, on a corpus linguistics and AI project e) hypothesize, and critically examine how corpus linguistics might be revolutionized by the maturity of AI technologies in the future

	Category B: Attributes for all-roundedness f) develop analytical reasoning, critical thinking, and problem - solving skills: • analytical reasoning: to think in a logical manner, supporting ideas with well-reasoned arguments and evidence • critical thinking: to evaluate information and evidence critically, the ability to recognize flaws or inconsistencies in an argument • problem solving: to understand a problem, explore plausible solutions, and pose the most appropriate decision/solution for the problem g) employ a range of strategies for effective communication and learning autonomously and collaboratively; and h) develop intellectual curiosity to work both independently and as part of a team
Subject Synopsis/ Indicative Syllabus	1. The theoretical assumptions and development of the field of corpus linguistics 2. Defining and understanding what a corpus is 3. Learning about the different types of corpora 4. Corpus software programs and tools, e.g. Antconc, ConcGram, Sketch Engine, WordSmith Tools, Wmatrix, etc. 5. Techniques to analyze corpus by: word, phrase, collocates, keyness, concordances, (including how AI might aid in analysis) 6. The mechanics of corpus creation: design, construction, and annotation (including how AI may aid in corpus creation) 7. Corpus-based, corpus-driven, and AI approaches to linguistic description and analysis 8. Applications of corpora, e.g. • tracking variation and change in the English language • producing dictionaries and other reference materials • studying aspects of linguistics, including morphology, lexis, grammar, lexico-grammar, semantics, literal and metaphorical meanings, discourse structure, pragmatics, and discourse • studying linguistic variation across modes (speaking and writing), registers (academic, business, social, scientific, legal, etc.) and genres (university textbooks, financial reports, conversation, laboratory reports, contracts, etc.) • building search engines, chatbots, smart assistants (Siri, Alexa), large language models, etc. 9. How to report on corpus linguistic findings 10. How corpus linguistics can both leverage and be leveraged by cutting edge technologies (e.g. artificial intelligence) for a wide range of outcomes

	11. How the field of corpus linguistics has been impacted (both positively and negatively) by the maturation of artificial intelligence							
Teaching/Learning Methodology	The teaching and learning approach is task-based, student-centered, interactive, and reflective.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	1. Paper	25%		✓	✓	✓	✓	✓
	2. Activities	15%	✓	✓	✓	✓	✓	✓
	3. Corpus Project	60%	✓	✓	✓	✓	✓	✓
	Total	100 %						
	The paper will give students an opportunity to critically discuss the effectiveness of adopting AI in corpus linguistic studies. The corpus project will invite students to use their knowledge of corpus linguistics and AI technologies to conduct an original research project. The activities, in the form of in-class worksheets, will consolidate student understanding of the content taught during the weekly lectures. All the assessed tasks allow the students to demonstrate how well they understand the subject knowledge, how effectively they can apply their knowledge of traditional and AI-assisted corpus processing and analysis, and how effectively they can conduct, report and critique corpus research.							
Student Study Effort Expected	Class contact:							
	▪ Lectures						39 Hrs.	
	▪ Other student study effort:							
	Private Study						58 Hrs.	
	▪ Assignment Preparation						29 Hrs.	
	▪ Total student study effort						126 Hrs.	
	Lectures						39 Hrs.	
Reading List and References	Recommended Readings							

	Baker, P., & Collins, L. (2023). Creating and analysing a multimodal corpus of news texts with Google Cloud Vision's automatic image tagger. <i>Applied Corpus Linguistics</i>, 3(1), 100043. Crawford, W., & Csomay, E. (2016). <i>Doing corpus linguistics</i>. Routledge. Crosthwaite, P., & Baisa, V. (2023). Generative AI and the end of corpus-assisted data-driven learning? Not so fast!. <i>Applied Corpus Linguistics</i>, 3(3), 100066. Curry, N., Baker, P., & Brookes, G. (2023). Generative AI for corpus approaches to discourse studies: a critical evaluation of ChatGPT. <i>Applied Corpus Linguistics</i>, 100082. Garside, R., Leech, G., & Sampson, G. (1987). <i>The computational analysis of English: A corpus-based approach</i>. Longman. Friginal, E., & Hardy, J. A. (Eds.). (2020). <i>The Routledge handbook of corpus approaches to discourse analysis</i>. Routledge. Lin, P. (2023). ChatGPT: Friend or foe (to corpus linguists)? <i>Applied Corpus Linguistics</i>, 3(3), 100065. O'Keeffe, A., & McCarthy, M. (Eds.). (2022). <i>The Routledge</i>
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Prepared in June 2024